Ensemble

↙

different Model + combine

Boosting ← Basic ← 1 2 3

Random Forest ←

Bagging

Bootstrap Aggregation

↓

Random sample → RS + CS → DT DT DT DT → Aggregate

DT

↙

High Variance and Low bias

Ensemble

Boosting

• combine weak learners

• High Variance — Bagging — Reduce variance

• Low Variance
• High Bias
} underfit

Avg → 15 LPA

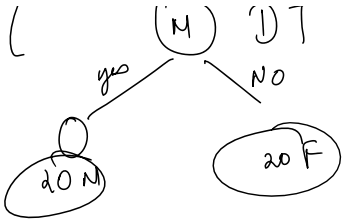
error ↑ High Bias
Variance ↓ Low

M → DT

yes / NO

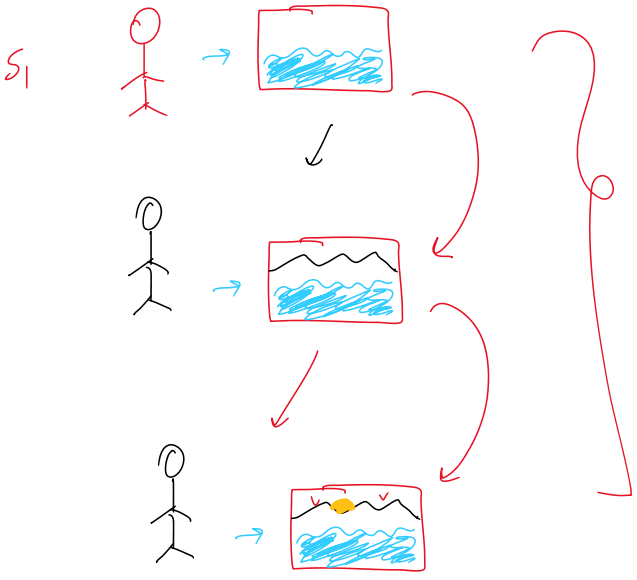
} Tree of 1 Depth

Stump



Tree of 1 Depth

Additive Combining



Additive

old + component → New

		Target	Mode Avg	error
C_1	C_2	y	$\hat{y}$	
	F	40	60	-20
	M	50	60	-10
	F	70	60	10
	M	80	60	20

Base Learner

weak

60

1 Model

$$\text{Target} = \hat{y} + \text{error}$$

$\hat{y}$: Avg model

$\text{error} = \hat{y}_{..} + \text{error}$

y Avg model + $\hat{y}_2$ + error

65 15

$y = h_0(x)$ + Error

$\hat{y}$

Predicted y

Avg y

60

weight

y ←

35 + 20 (error)

Base

35 + 10 + 10 (error)

x_0 x_1

Quiz 2

— high bias — Low variance

→ Mean — high bias Low variance.

→ Random → high bias high variance

Perfect fit low bias ✗

error

high bias

1

error

y = Avg + error

$h_0(x)$

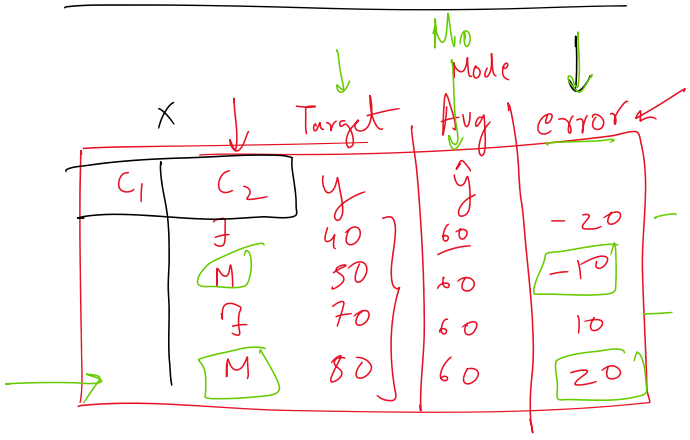
$\hat{y}$

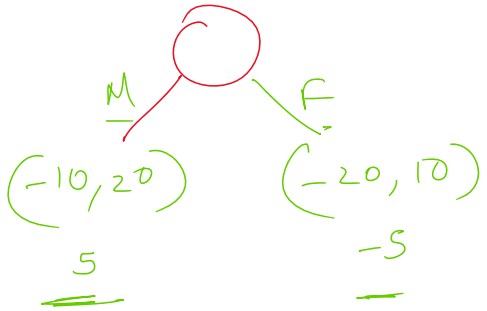
Predicting error

↓

DT

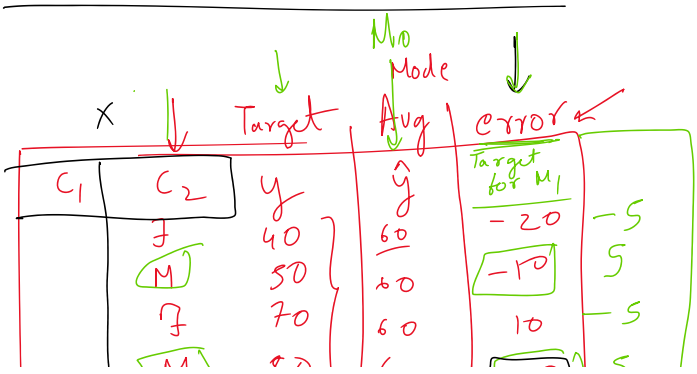
M_0



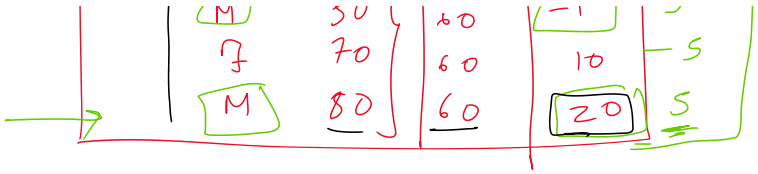


error

$y = 60 + 20$



$h_0(x) + h_1(x) + e$



$y - y_{Pred}$

error

$y = 60 + 20$

error

$y = 60 + 5 + 15$

$y = h_0(x) + h_1(x) \dots\dots h_m(x) + \text{error}$

$y = h_0(x) + \alpha h_1(x) \dots\dots \alpha h_m(x) + \text{error}$

control Parameter

0.01×100

$\downarrow$

0.1

0.01×100

1

M_0 Avg → error

M_1 → Predict

C

Low weight

$y = m_1 x_1 + m_2 x_2$

less

less ↓ x impact on y is less

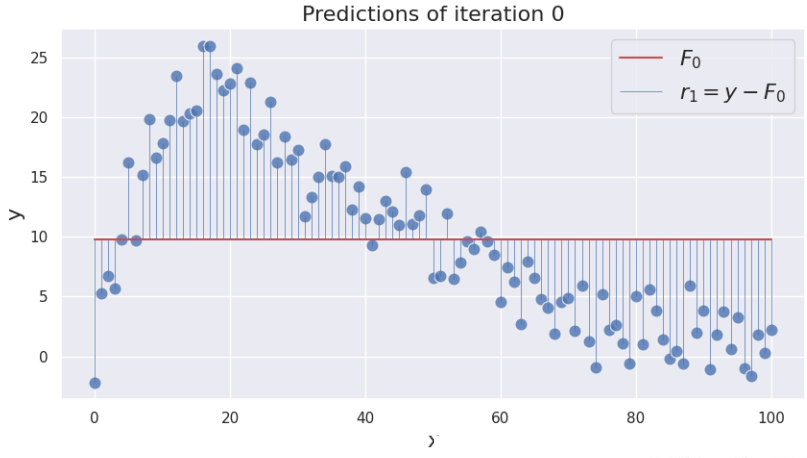
Low

$F = M_0 + \alpha M_1 + M_2 + \text{error}$

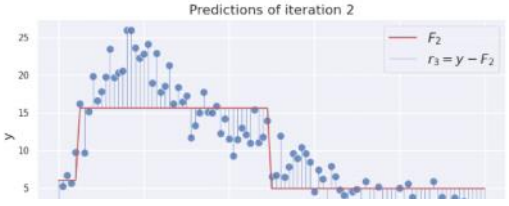
Low

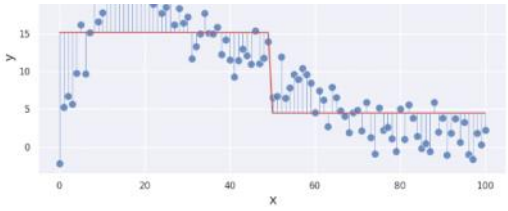
0.0001

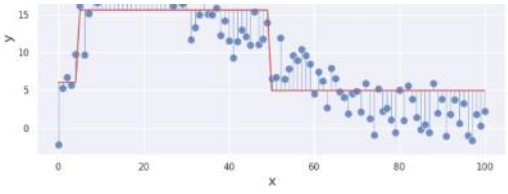
$$\begin{cases} y = h_0 x + h_1(x) + h_2(x) \\ \downarrow \quad \downarrow \quad \downarrow \\ M_0 \quad M_1 \quad M_2 \end{cases}$$ Sequential

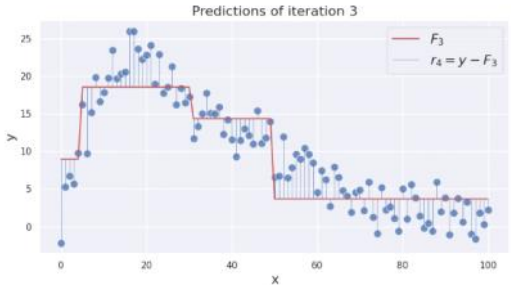


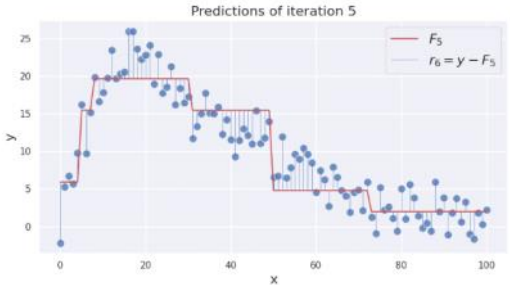




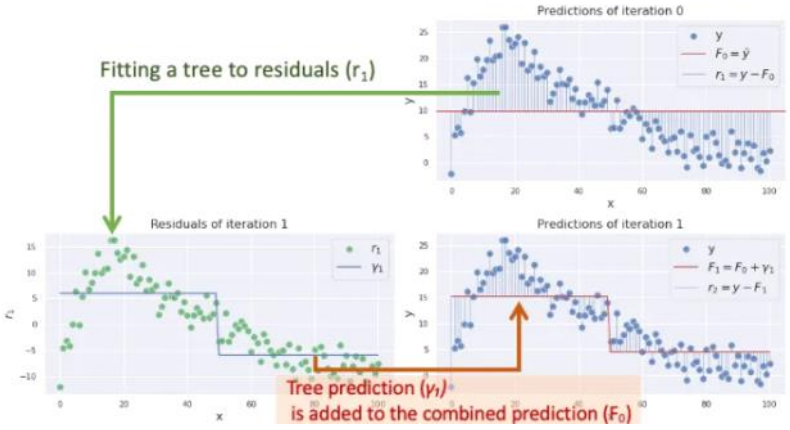




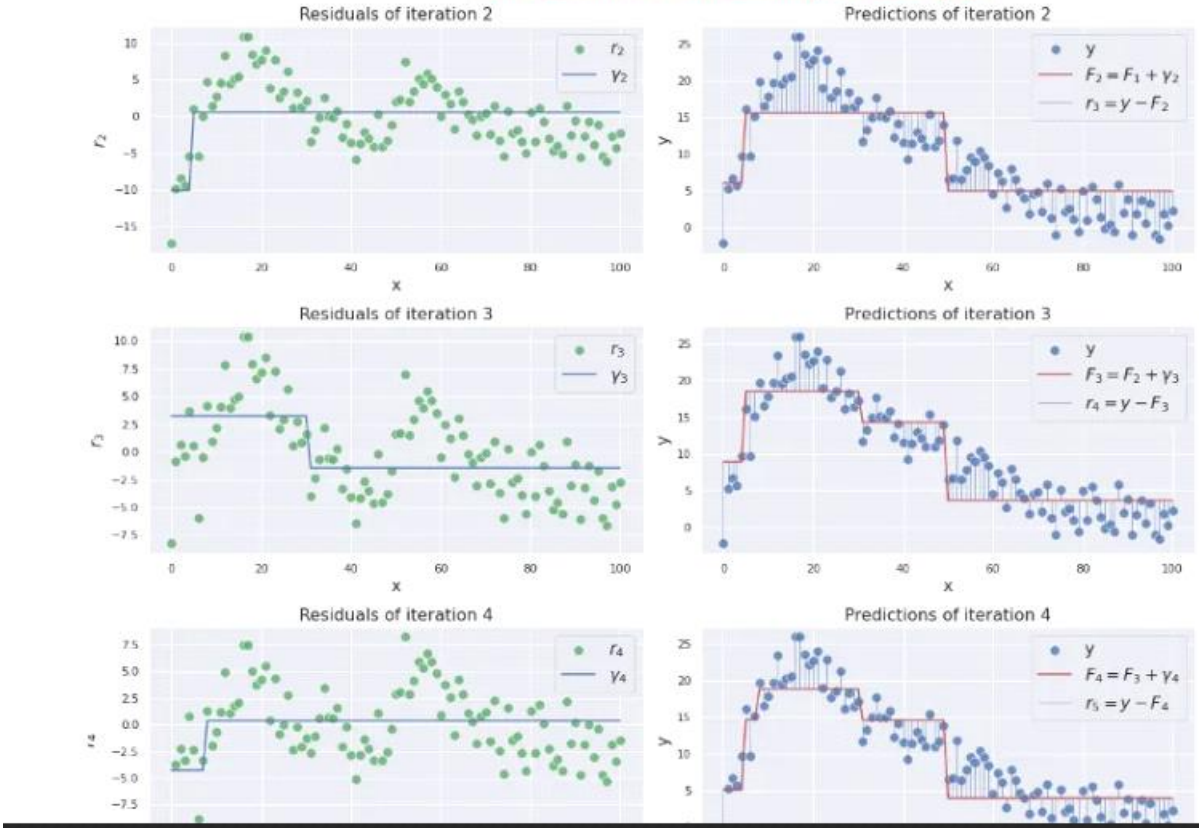




The whole process along with the reduction in residual looks as follows :



x is added to the combined prediction (F_0)



$$L(y, \hat{y}) = (y - \hat{y})^2$$ Squared

$$\frac{\partial L}{\partial \hat{y}} = 2(y - \hat{y}) \times \frac{\partial (y - \hat{y})}{\partial \hat{y}}$$

$$\frac{\partial L}{\partial \hat{y}} = -2(y - \hat{y})$$

$$-\frac{\partial L}{\partial \hat{y}} = 2\boxed{(y - \hat{y})}$$ Loss

error

Pseudo error

Gradient Boosting Decision Trees (GBDT) is an ensemble learning technique that combines multiple decision trees to create a powerful predictive model. The key idea behind GBDT is to sequentially build trees, with each new tree attempting to correct the errors made by the previous ones. Pseudo-residuals play a crucial role in this process.

Here's a simplified explanation of how pseudo-residuals are used in GBDT:

Initial Prediction: GBDT starts by making an initial prediction for each sample in the dataset. This initial prediction could be the mean (for regression problems) or the most frequent class (for classification problems). These initial predictions are often referred to as the "pseudo-residuals" at the beginning because they represent the errors between the actual target values and the initial predictions.

Building Trees to Correct Errors: After making initial predictions, GBDT starts building decision trees to correct these errors. Each decision tree is trained to predict the residuals, which are the differences between the actual target values and the current predictions made by the ensemble of trees built so far.

Updating Predictions: Once a decision tree is built, it predicts the residuals for each sample in the dataset. These predicted residuals are then added to the current predictions (or pseudo-residuals) to update the

ensemble's predictions. The ensemble's predictions are gradually refined with each new tree added to the model.

Sequential Learning: GBDT adds trees sequentially, with each new tree focusing on reducing the errors (pseudo-residuals) that were not adequately addressed by the previous trees. This sequential learning process continues until a predefined number of trees are built or until a certain level of performance is achieved.

Here's a step-by-step example to illustrate the concept:

Let's say we have a regression problem where we want to predict housing prices based on features like square footage, number of bedrooms, and location.

We start by making an initial prediction for each house in the dataset. This could be the mean of all housing prices in the dataset.

We calculate the residuals, which are the differences between the actual housing prices and the initial predictions. These residuals represent the errors in our initial predictions.

We build a decision tree to predict these residuals. The decision tree is trained to find patterns in the features that are associated with the residuals.

The decision tree predicts the residuals for each house in the dataset.

We update our predictions by adding the predicted residuals to our initial predictions. This gives us updated predictions that should be closer to the actual housing prices.

We repeat steps 3-5, building new trees to predict the remaining errors (pseudo-residuals) in our predictions. Each new tree focuses on correcting the errors that were not adequately addressed by the previous trees.

Finally, we combine the predictions from all the trees to get the final prediction for each house.

In summary, pseudo-residuals in GBDT represent the errors between the actual target values and the current predictions made by the ensemble of trees. By sequentially building trees to predict these residuals, GBDT gradually improves the accuracy of its predictions and creates a powerful predictive model.

